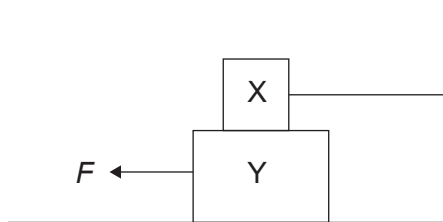


PAPER 1 / QUESTION 54

25 April 2024 / TZ2 (file label) / HL Paper 1 / original question 3 / 2224-6707

54. A block X of weight 10 N is stacked on a block Y of weight 20 N. Block X is fixed to a wall with a light string. The coefficients of static friction between the blocks and between block Y and the ground are both 0.2.



What is the value of the minimum force F required to move block Y and what is the tension T in the string immediately before block Y begins to move?

	Minimum force F/N	Tension in string T/N
A.	8	2
B.	8	6
C.	6	2
D.	6	6